DESIGN AND KINEMATIC SIMULATION OF A 3-DOF MANIPULATOR

(ROBOTIC ARM)

Internship Project Report

ABSTRACT

This project presents the design, visualization, and kinematic simulation of a three-degree-of-freedom (3-DOF) SCARA robotic manipulator. The work follows a sequential workflow beginning with 3D modeling of the robot structure, followed by animation for visual verification of joint motion, and concluding with an independent kinematic simulation implemented in MATLAB. The SCARA robot consists of two revolute joints operating in the horizontal plane and a prismatic joint providing vertical motion.

The 3D model was initially developed and animated to study the mechanical configuration and joint behaviour. Separately, a mathematical model of the robot was created in MATLAB to analyse forward and inverse kinematics, trajectory motion, and pick-and-place operations. An interactive graphical user interface was implemented to allow manual and automatic control of the robot joints, demonstrating smooth and realistic motion.

The results validate the correctness of the kinematic model and show consistency between visual animation and mathematical simulation. The project provides a clear understanding of SCARA robot motion and serves as a foundation for future hardware implementation using actuators and microcontroller-based control systems.

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CHAPTER 1

INTRODUCTION

Robotic manipulators have become an essential part of modern automation systems due to their ability to perform repetitive tasks with high accuracy, speed, and reliability. Industries such as manufacturing, electronics assembly, packaging, and material handling extensively use robots to improve productivity and reduce human effort. Among the various types of industrial robots, the **SCARA (Selective Compliance Assembly Robot Arm)** configuration is widely preferred for planar operations that require precision and rapid motion.

A SCARA robot typically consists of two revolute joints that operate in the horizontal plane and a prismatic joint that provides vertical motion along the Z-axis. This structure allows the robot to maintain rigidity in the vertical direction while being compliant in the horizontal plane, making it highly suitable for pick-and-place operations and assembly tasks. Due to its simple mechanical structure and ease of control, the SCARA robot is commonly used as a learning platform for understanding robotic kinematics and motion planning.

The study of robot kinematics plays a crucial role in analyzing and controlling robotic motion. Forward kinematics is used to determine the position of the end effector based on given joint variables, while inverse kinematics is used to compute the required joint variables for a desired end-effector position. Accurate kinematic modeling enables effective trajectory planning, workspace analysis, and motion simulation of robotic systems before physical implementation.

This project focuses on the design, visualization, and kinematic simulation of a three-degree-of-freedom (3-DOF) SCARA robotic manipulator. A sequential workflow is followed, beginning with three-dimensional modeling of the robot structure, followed by animation to visually verify joint movements, and concluding with an independent kinematic simulation implemented in MATLAB. The visual modeling and animation help in understanding the mechanical configuration, while the MATLAB simulation provides a mathematical analysis of robot motion using forward and inverse kinematics.

The primary objective of this project is to develop a clear understanding of SCARA robot structure and kinematic behavior through simulation. The work also demonstrates trajectory-based motion and pick-and-place operations using an interactive simulation environment. Additionally, the project establishes a strong foundation for future extension toward hardware implementation using actuators and microcontroller-based control systems.

CHAPTER 2

PROJECT WORKFLOW, TOOLS USED, AND IMPLEMENTATION

This chapter describes the complete workflow followed in the project along with the specific tools used at each stage. The development of the SCARA robotic manipulator was carried out in a sequential manner, beginning with 3D design, followed by animation for motion visualization, and finally an independent kinematic simulation using MATLAB. Each tool was selected based on its suitability for a particular task in the development process.

2.1 Tools Used

The following tools were used during the implementation of the project:

- **Tinkercad** – Used for designing and modeling the individual components of the SCARA robot
- **Blender** – Used for importing components, joint alignment, and animation of robot motion
- **MATLAB** – Used for independent kinematic modeling, simulation, and trajectory-based motion analysis

Each tool served a distinct purpose and was not used interchangeably.

2.2 3D Design of the SCARA Robot Using Tinkercad

The first stage of the project involved designing the SCARA robotic manipulator in **Tinkercad**. Individual components such as the base, central column, robotic links, and end-effector structure were modeled separately. Designing the parts individually allowed better control over dimensions and joint placement.

Once all components were created, they were assembled virtually to visualize the complete SCARA robot structure. This stage helped in understanding the mechanical configuration and spatial arrangement of the robot before proceeding to animation.

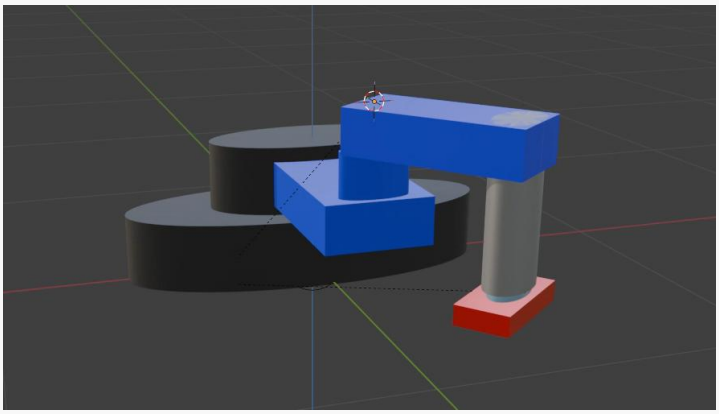
After completing the design, the individual parts were exported from Tinkercad in a compatible 3D format for further processing.

2.3 Importing, Alignment, and Animation Using Blender

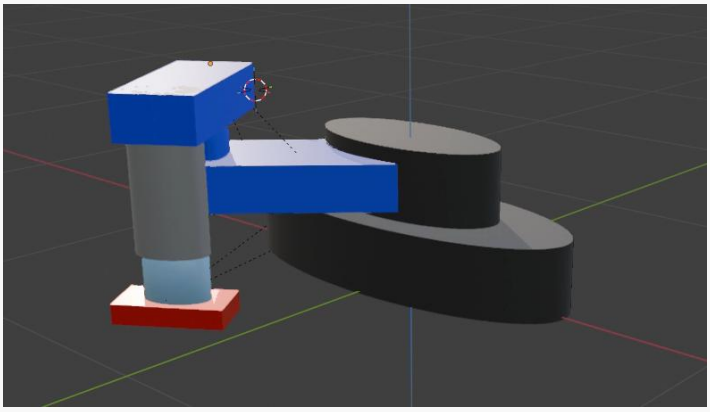
The exported components from Tinkercad were imported **individually** into **Blender** for animation and motion visualization. Importing parts separately was necessary to correctly define joint behavior and relative motion between components.

In Blender, the following steps were carried out:

- Setting correct **origins** for each component to represent joint axes
- Aligning all parts accurately to reconstruct the complete robot structure
- Assigning rotational motion to the revolute joints
- Assigning translational motion to the prismatic joint
- Verifying joint limits and motion directions



After joint setup, animation keyframes were applied to simulate realistic SCARA robot motion. The animation was used to visually verify joint behavior such as rotation, extension, and retraction. This stage focused purely on **visual and mechanical motion representation** and did not involve any mathematical or kinematic calculations.



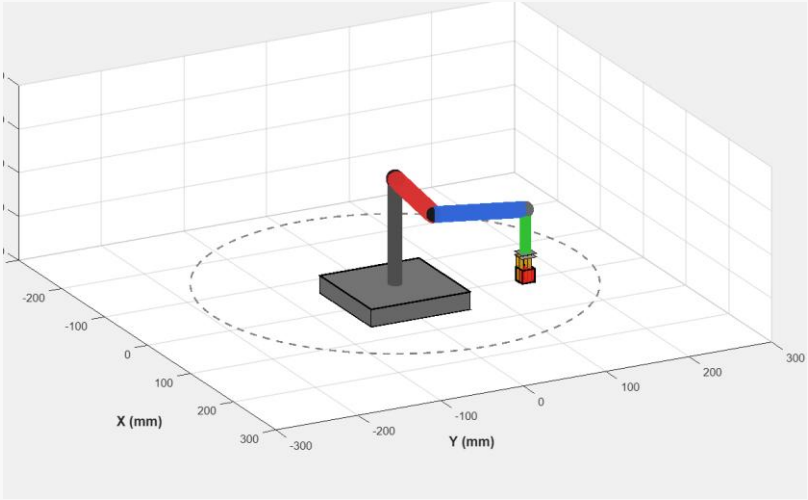
The final animation was recorded to document the robot motion and to serve as visual evidence of correct joint operation.

2.4 Independent Kinematic Modeling and Simulation Using MATLAB

In the final stage of the project, an **independent model** of the SCARA robot was developed using **MATLAB**. The MATLAB model was not created by importing geometry from Tinkercad or Blender. Instead, the robot was defined mathematically using link lengths, joint variables, and coordinate relationships.

The MATLAB implementation included:

- Definition of robot geometric parameters
- Forward kinematics to compute end-effector position from joint values
- Inverse kinematics to determine joint angles for desired positions
- Trajectory generation using smooth interpolation
- Simulation of pick-and-place operations
- Interactive control using a graphical user interface



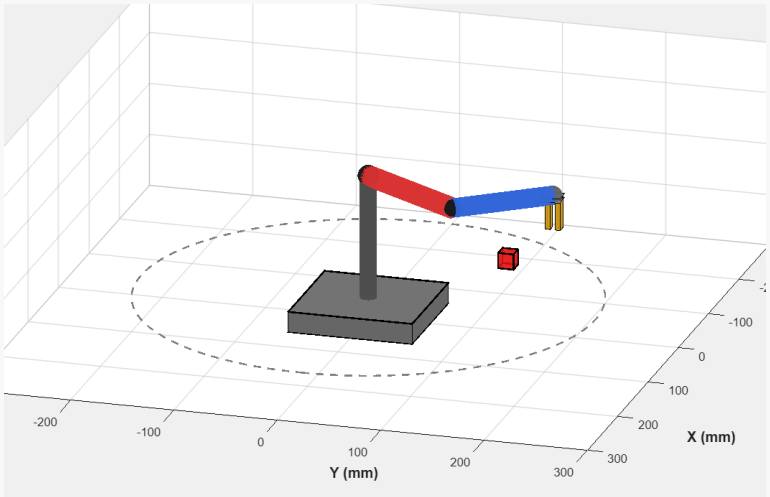
This simulation enabled analysis of robot motion based on kinematic principles and allowed verification of trajectory behavior and workspace limits. The MATLAB model complemented the Blender animation by providing mathematical validation of robot motion.

2.5 Workflow Summary

The complete workflow of the project can be summarized as follows:

1. Design of individual robot components using **Tinkercad**
2. Export of components and individual import into **Blender**
3. Alignment, joint setup, and animation for motion visualization
4. Independent kinematic modeling and simulation using **MATLAB**

This structured approach ensured clear separation between mechanical visualization and mathematical analysis while providing a comprehensive understanding of the SCARA robot system.


-200
-100
0
100
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Y (mm)
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X (mm)

CHAPTER 3

MATLAB-BASED KINEMATIC MODELING AND SIMULATION

This chapter presents the MATLAB-based kinematic modeling and simulation of the 3-DOF SCARA robot. The robot is modeled independently using mathematical relationships, and forward kinematics, inverse kinematics, and trajectory-based motion are implemented through MATLAB code.

3.1 Definition of Robot Parameters

The geometric parameters of the SCARA robot such as link lengths and base dimensions are defined to model the robot structure mathematically.

```
robot.L1 = 120;            % Link 1 length
robot.L2 = 100;            % Link 2 length
robot.base_height = 20;
robot.column_height = 120;
```

3.2 Joint Variables Initialization

Joint variables corresponding to two revolute joints and one prismatic joint are initialized. The prismatic joint is defined to move downward along the Z-axis.

```
joint.theta1 = 0;          % Base rotation
joint.theta2 = pi/4;       % Elbow rotation
joint.d3 = 0;              % Vertical extension (downward)
```

3.3 Forward Kinematic Modeling

Forward kinematics is used to compute the end-effector position from the joint variables. The X and Y coordinates are obtained from the planar revolute joints, while the Z coordinate is controlled by the prismatic joint.

```
x2 = robot.L1 * cos(joint.theta1) + ...
     robot.L2 * cos(joint.theta1 + joint.theta2);

y2 = robot.L1 * sin(joint.theta1) + ...
     robot.L2 * sin(joint.theta1 + joint.theta2);

z_platform = robot.base_height + robot.column_height;
z_gripper = z_platform - joint.d3;
```

3.4 Inverse Kinematic Modeling

Inverse kinematics is implemented to calculate joint angles for a desired end-effector position during pick-and-place operations. The solution uses the law of cosines and includes reachability checks.

```
pick_dist = sqrt(pick_x^2 + pick_y^2);

cos_theta2 = (pick_dist^2 - robot.L1^2 - robot.L2^2) / ...
             (2 * robot.L1 * robot.L2);

theta2_pick = acos(cos_theta2);

alpha = atan2(pick_y, pick_x);
beta = atan2(robot.L2 * sin(theta2_pick), ...
             robot.L1 + robot.L2 * cos(theta2_pick));

theta1_pick = alpha - beta;
```

3.5 Trajectory Generation

Smooth motion between joint configurations is achieved using linear interpolation over multiple steps.

```
current = start + t * (target - start);

joint.theta1 = current(1);
joint.theta2 = current(2);
joint.d3     = current(3);
```

3.6 Interactive Control and Simulation

An interactive graphical user interface (GUI) is implemented to allow real-time manual control of joints and execution of automated motion sequences.

3.7 Pick-and-Place Operation

A complete pick-and-place operation is simulated by combining inverse kinematics and trajectory control, enabling the robot to grasp, move, and place an object.

CHAPTER 4

RESULTS AND OBSERVATIONS

This chapter presents the results obtained from the 3D animation and MATLAB-based kinematic simulation of the SCARA robotic manipulator. The observations are based on visual verification, joint behavior, and trajectory execution.

4.1 Results from 3D Modeling and Animation

- The SCARA robot structure was successfully modeled and assembled using individual components.
- Joint origins and alignment in Blender resulted in correct rotational and translational motion.
- The animation clearly demonstrated planar motion of the revolute joints and vertical motion of the prismatic joint.
- Visual inspection confirmed smooth joint movements without misalignment or unrealistic behavior.

4.2 Results from MATLAB Kinematic Simulation

- Forward kinematics correctly computed the end-effector position for different joint configurations.
- Inverse kinematics successfully calculated joint angles for reachable target positions.

- The prismatic joint motion was observed to extend downward correctly, matching the SCARA configuration.
- Smooth trajectory motion was achieved through interpolation, avoiding abrupt joint movements.

4.3 Pick-and-Place Operation Results

- The simulated pick-and-place operation was executed successfully using inverse kinematics.
- The robot was able to approach the object, lower the end effector, grasp the object, and place it at a target location.
- Object motion remained synchronized with the end effector during transport.
- Reachability checks prevented invalid target positions outside the robot workspace.

4.4 Observations

- The visual animation and MATLAB simulation showed consistent robot behavior.
- Separating animation and kinematic modeling improved clarity and accuracy of analysis.
- The simulation effectively demonstrated the working principles of a 3-DOF SCARA robot.
- The developed model is suitable for extension toward real-time control and hardware implementation.

CHAPTER 5

HARDWARE FEASIBILITY AND FUTURE SCOPE

This chapter briefly discusses the feasibility of implementing the proposed SCARA robot model in hardware and possible future extensions of the project.

5.1 Hardware Feasibility

The developed SCARA robot model is suitable for hardware implementation due to its simple and well-defined joint structure. The two revolute joints can be implemented using servo or stepper motors, while the prismatic joint can be realized using a linear actuator or lead screw mechanism. A microcontroller such as an Arduino can be used to generate control signals for joint actuation.

The joint variables obtained from the kinematic simulation can be mapped to motor commands, enabling real-time control of the physical robot. Since the kinematic model follows standard SCARA configuration, the transition from simulation to hardware is feasible with appropriate mechanical design and control integration.

5.2 Future Scope

The project can be extended by incorporating dynamic analysis, closed-loop control using sensors, and collision detection. Future work may also include physical fabrication of the robot and real-time control using embedded systems for industrial pick-and-place applications.

CONCLUSION

This project successfully demonstrates the design, animation, and kinematic simulation of a 3-DOF SCARA robotic manipulator. The robot was modeled and animated to verify joint motion, and an independent MATLAB simulation was used to implement forward and inverse kinematics along with trajectory-based motion. The results confirm correct robot behavior and effective execution of pick-and-place operations. The developed model provides a basic foundation for future extension toward hardware implementation.

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